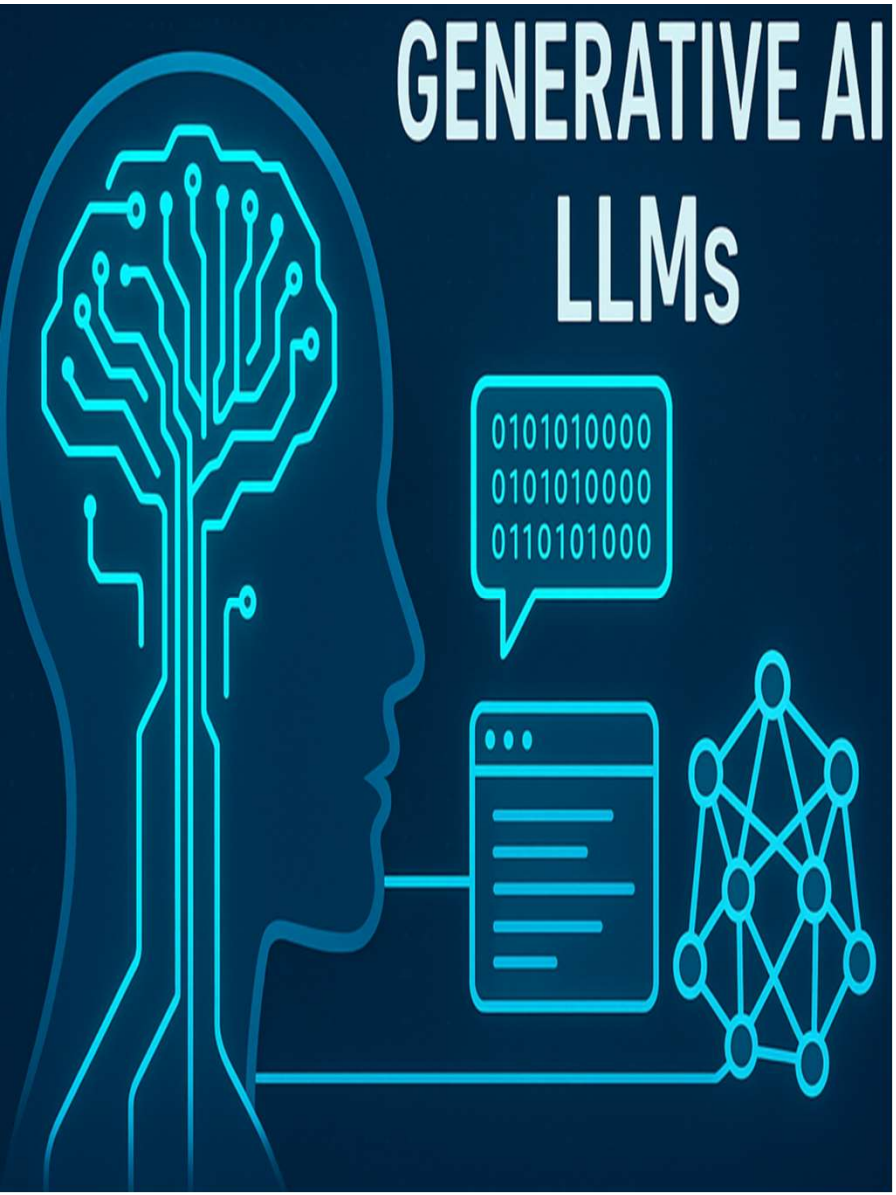


9 – Generative AI, LLMs, and Agentics

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Curricular Unit: Artificial Intelligence

Programme: BSc in Informatics Engineering (ISEP)

Game Changer Technology: Transformers

In 2017 Indian researcher Ashish Vaswani from Google Brain publishes his work "Attention is all you need" (https://papers.nips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf) proposing the ANN/DL Transformer architecture, and emphasising the role of Attention Mechanism

Attention Is All You Need

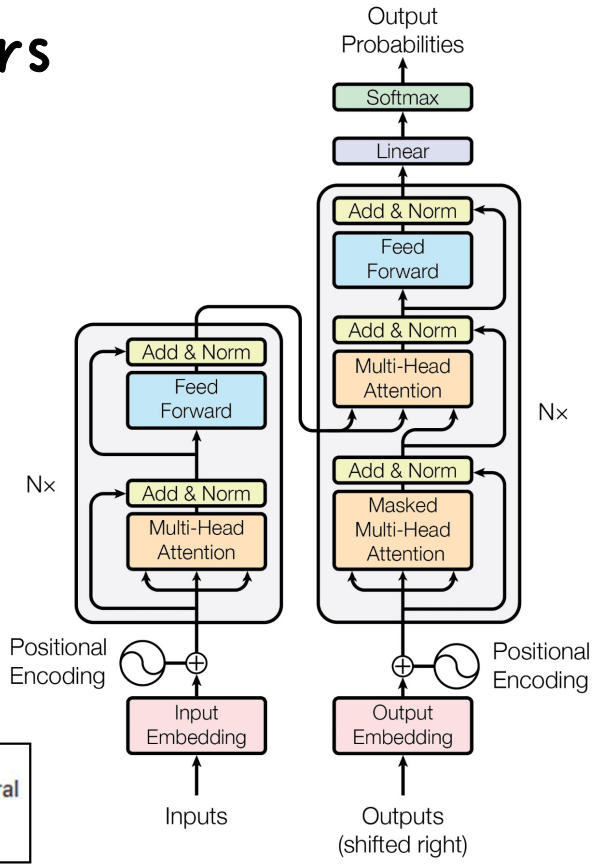
Ashish Vaswani* Google Brain avaswani@google.com
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Jakob Uszkoreit* Google Research usz@google.com
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

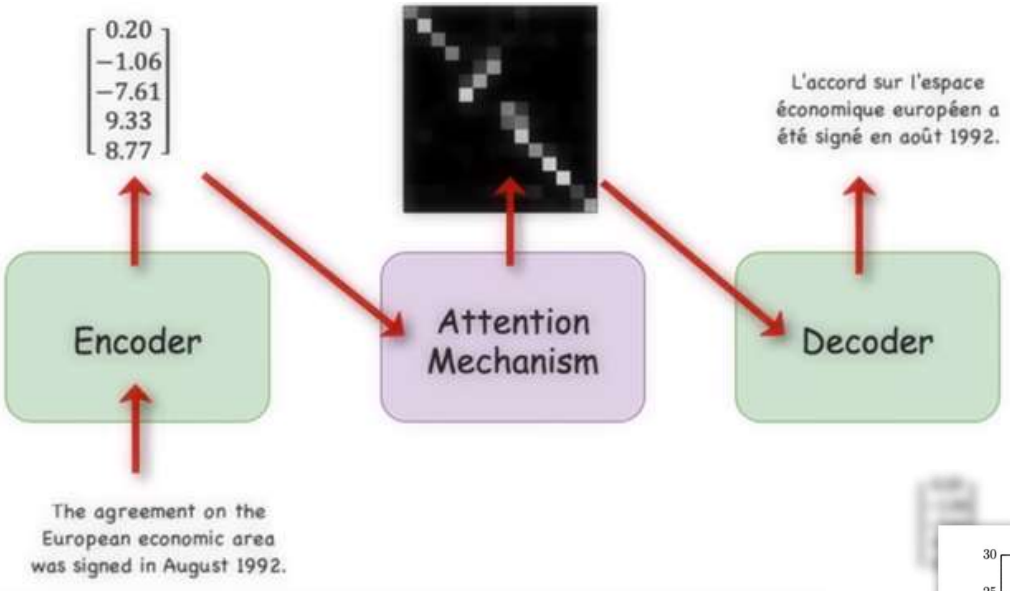


Attention is all you need
A Vaswani, N Shazeer, N Parmar, J Uszkoreit, L Jones... - Advances in neural information processing systems, 2017
Cited by 206457 Related articles All 70 versions

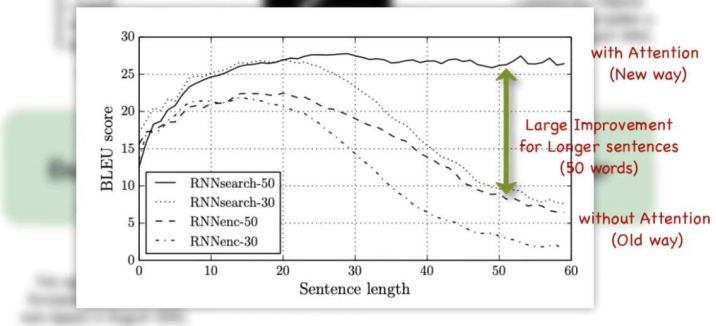


The Transformer architecture has been used for several applications of Natural Language Processing (eg in Machine Translation) and was the base of the great success of Generative AI, with a revolutionary product (ChatGPT) emerging in late 2022.

Attention Mechanism

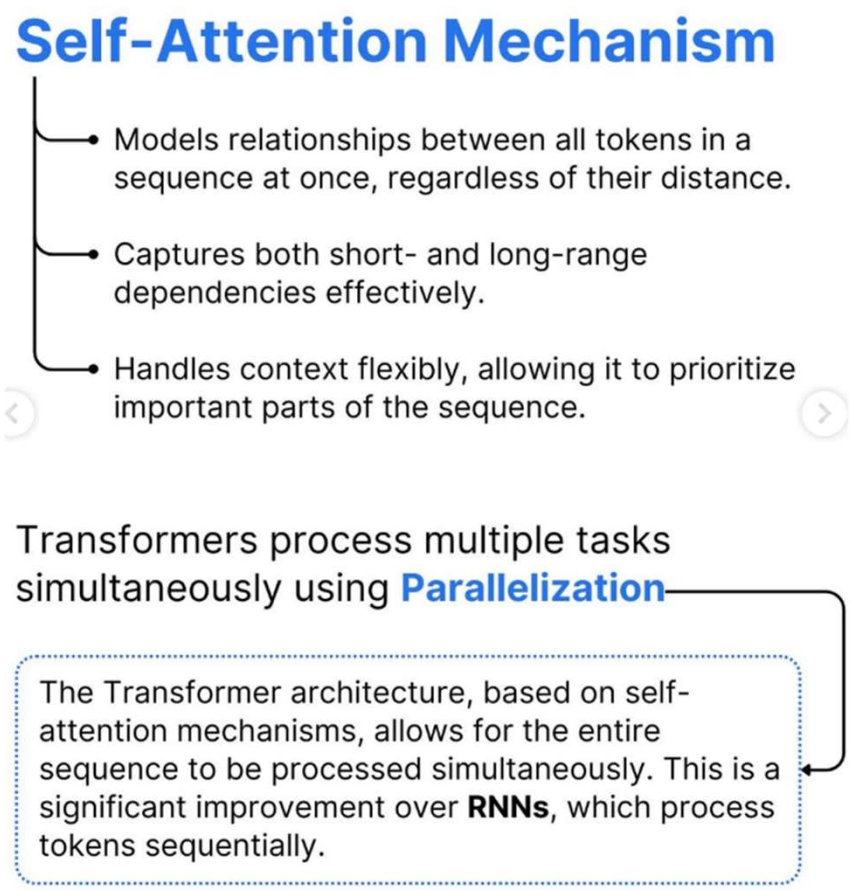


The Attention Mechanism focuses on a more important part resulting from the input in this case in words that are more important



Source: Neural Machine Translation by jointly learning to align & translate (Bahdanau et al., 2016)

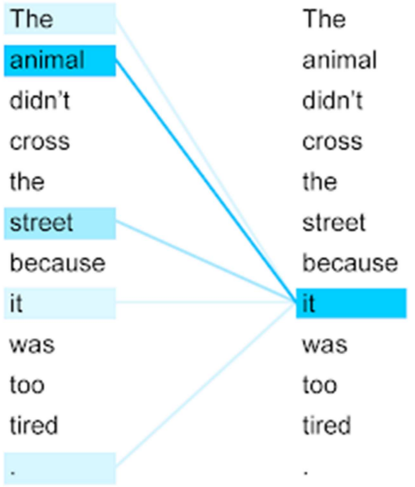
Self-Attention Mechanism of Transformers



Self-Attention Visualization

The encoder self-attention distribution for the word “it”:

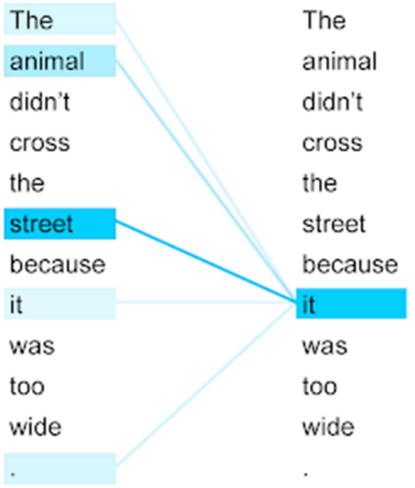
- associated to the animal due to “tired”
- or associated to the street due to “wide”



https://ai.googleblog.com/2017/08/transformer-novel-neural-network.html

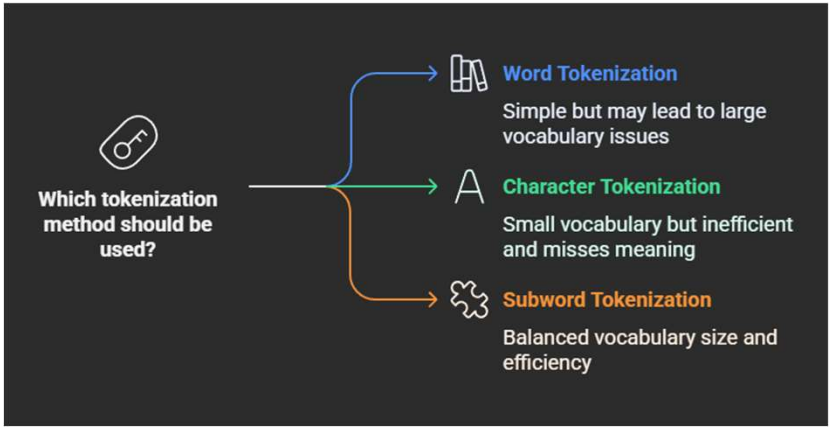
“I poured water from the bottle into the **cup** until **it** was **full**.”
it → cup
“I poured water from the **bottle** into the cup until **it** was **empty**.”
it → bottle

Another example



Tokens

In generative AI, tokens are the fundamental units of data that AI models use to process and generate language. They are created through a process called tokenization, which breaks text down into smaller pieces—like words, parts of words, or punctuation—so the model can understand and manipulate it. The concept of tokens is also tied to the AI's context window, which is the maximum number of tokens it can consider at one time for a given task.



Tokens are words, character sets, or combinations of words and punctuation that are generated by large language models (LLMs) when they decompose text. Tokenization is the first step in training. The LLM analyzes the semantic relationships between tokens, such as how commonly they're used together or whether they're used in similar contexts. After training, the LLM uses those patterns and relationships to generate a sequence of output tokens based on the input sequence.

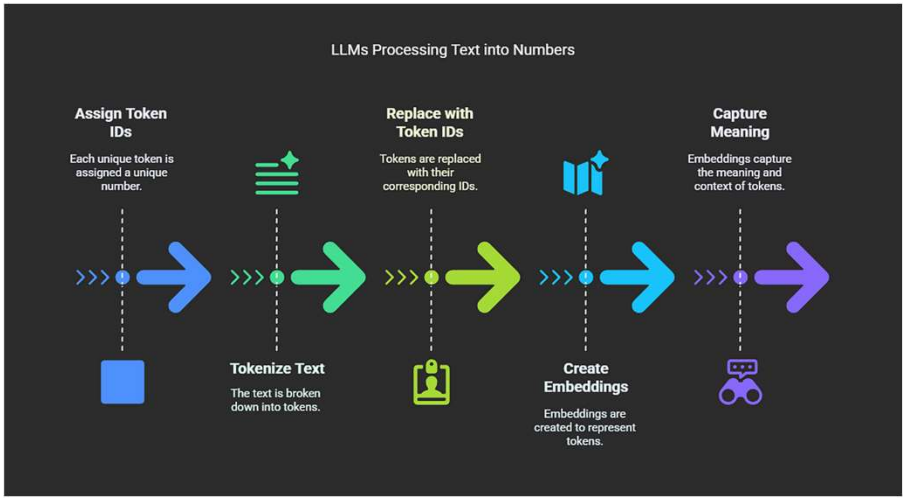
https://learn.microsoft.com/en-us/dotnet/ai/conceptual/understanding-tokens

Token/Word Vectorization

Artificial Neural Network and Deep Learning models work on numbers

For that reason we need to convert tokens/words in vectors

When we give a semantic relationship for words we refer the term embedding



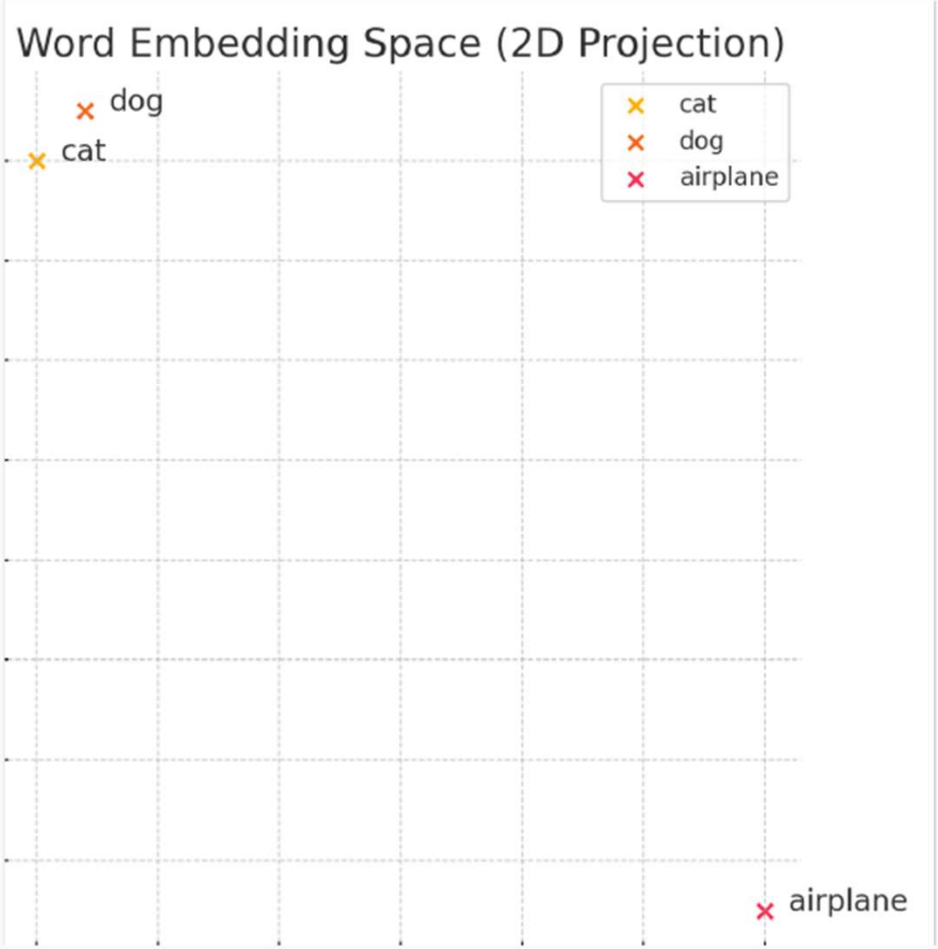
Word Vectorization

- Consider three words: **cat**, **dog**, and **airplane**
- **Cat** and **dog** have **similar vectors**, while **airplane** is far away, showing different meanings.

Word	Embedding (simplified)
cat	[0.21, 0.55, 0.63]
dog	[0.23, 0.58, 0.60]
airplane	[0.91, 0.12, 0.05]

Word Vectorization

- Embeddings represent words as dense vectors in a continuous space.
- Words with similar meanings (e.g., ”cat“ and ”dog“) are close together.
- Words with different meanings (e.g., ”airplane“) are far apart.



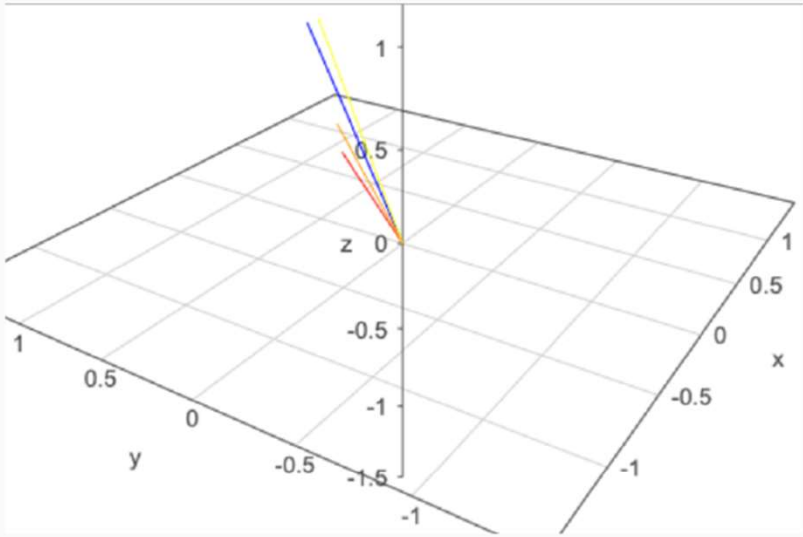
Word Vectorization

Word Analogy Example

A well-known example of how embeddings can capture semantic relations, including analogies.

”king“ - ”man“ + ”woman“ ≃ ”queen“

Word	Embedding (simplified)
king	[0.6, 0.9, 0.7]
man	[0.2, 0.5, 0.3]
woman	[0.3, 0.6, 0.4]
queen	[0.7, 1.0, 0.8]



$king - man + woman = [0.6 - 0.2 + 0.3, 0.9 - 0.5 + 0.6, 0.7 - 0.3 + 0.4] \simeq [0,7, 1.0, 0.8] \simeq queen$

Generative AI and Large Language Models

Generative AI (GenAI) is a deep learning model capable of generating various forms of content, such as text, video, images, and even music, depending on the training data it receives. Rather than taking training data and using it to make predictions like previous AI models, GenAI instead uses it to learn to create its own unique, similar data. For example, you could give a GenAI model a set of data consisting of pictures of cats. The model would then learn to recognize patterns and the distribution of pixels throughout the images to be able to produce its own pictures of cats

A Large Language Model (LLM) is another type of artificial intelligence powered by deep learning that can generate human language. It allows the model to write text, translate different languages, answer questions, and perform other tasks related to language. LLMs work by analyzing massive amounts of training data and learning to predict what word will come next in a sentence by assigning words a probability score based on the surrounding context. Eventually, LLMs gain the knowledge to pick up on these patterns accurately and generate relevant text.

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Advantages and Disadvantages of GenAI

Advantages of GenAI

Choosing to implement GenAI systems can offer your organization several advantages. For businesses, automating routine processes can lead to significant increases in productivity by freeing employees to focus on other work. Additionally, GenAI can improve customer service by providing access to chatbots at any hour of the day. It can also help develop more personalized experiences and product offerings.

Disadvantages of GenAI

Although GenAI certainly has its benefits, some challenges and potential downsides also exist. One such challenge is the ethical concerns raised by GenAI. Sometimes, people may use GenAI to create fake content and misinform the public. Also, GenAI models may contain sensitive data, making it important to encrypt data and follow data privacy regulations.

Coursera

Advantages and Disadvantages of LLMs

Advantages of LLMs

A significant advantage of utilizing LLMs is time-saving, since you can automate several different processes. Another benefit is their flexibility: You can train them to perform a wide range of tasks. They can provide innovative solutions to problems, offering a new perspective on challenges.

Disadvantages of LLMs

When working with LLMs, it's important to consider some of their challenges. For example, to use LLMs effectively and generate usable results, you must know how to create the correct prompts. Additionally, the quality of your LLM output can vary depending on the training data it receives and how you use it.

Coursera

Some other Disadvantages of LLMs

Hallucinations

When LLMs don't know an answer, they often confidently make up information that sounds believable rather than admitting ignorance. This can lead to the presentation of fictional details or events as facts

Limited Reasoning

Despite appearing intelligent, LLMs often struggle with basic math and complex problem-solving. They excel at understanding and generating language, but falter when faced with multi-step problems that require logical reasoning

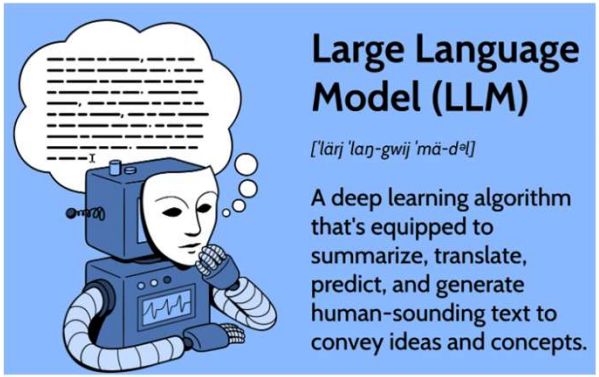
Knowledge Constraints

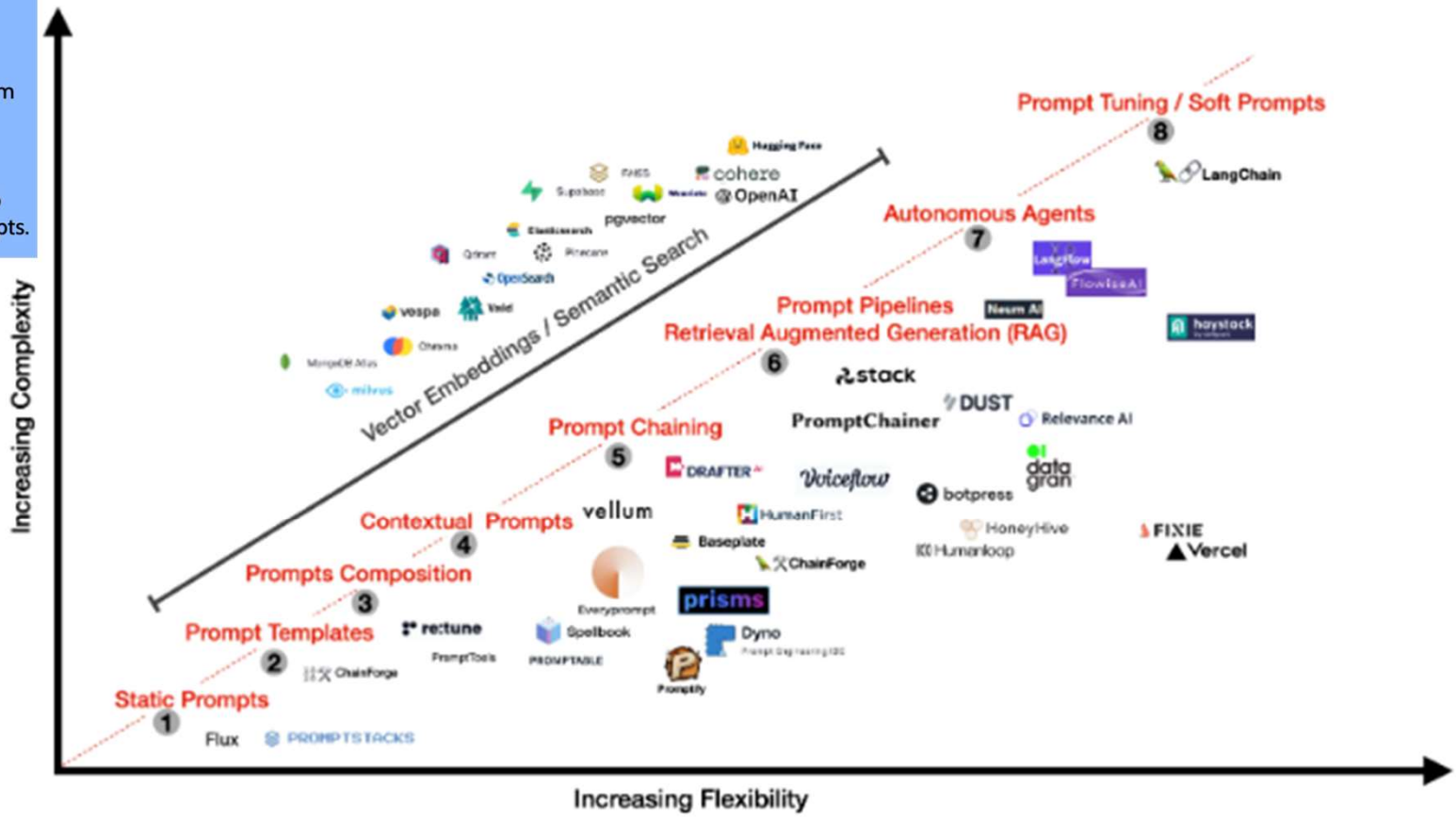
LLMs are trained on historical data and lack real-time information. They have no memory of previous conversations unless explicitly reminded, and they can reflect biases present in their training data

Large Computational Resources

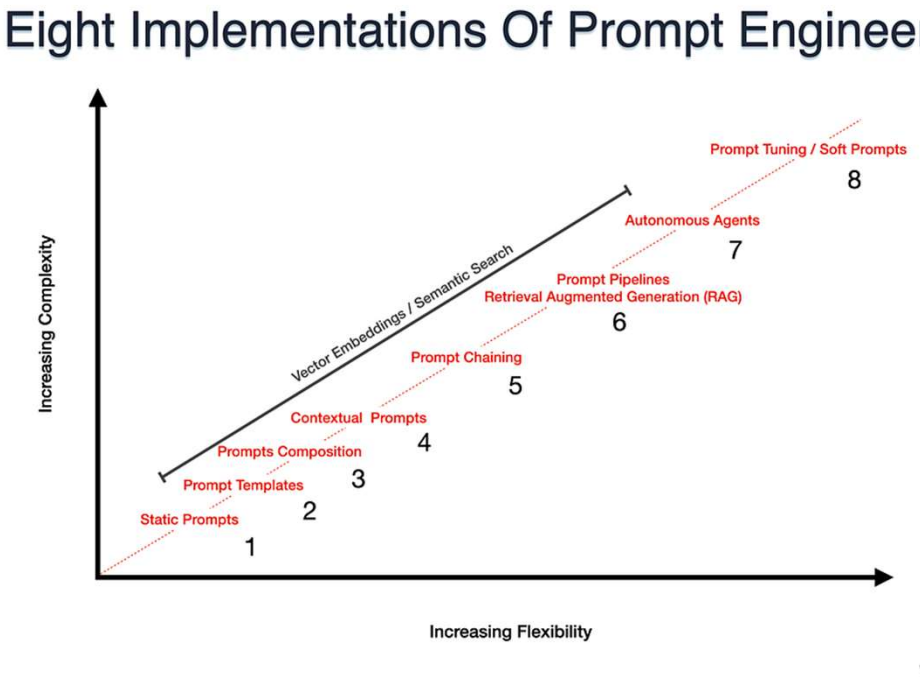
Both for training and inference

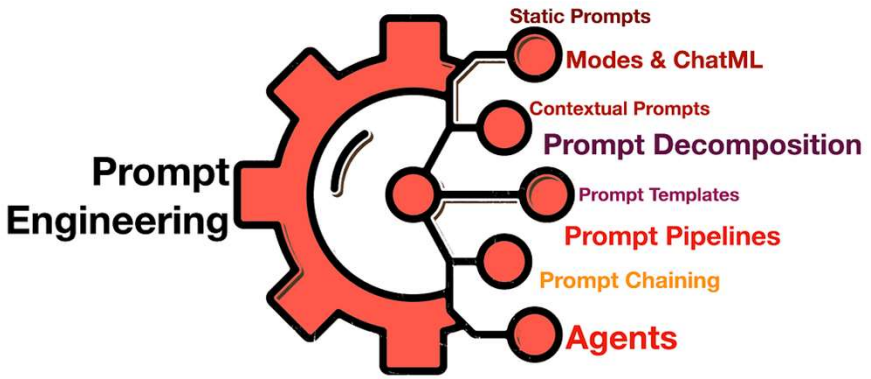
Large Language Models (LLM)





Prompt Engineering





https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350

Static Prompts

Static Prompts

The generative capabilities of LLMs are one of their key features which can be leveraged; with Prompt Engineering as the avenue with which data is presented and hence dictate the way the LLM executes on the data.

Prompts can follow a zero, single or few shot learning approach. The generative capabilities of LLMs are greatly enhanced by following a one-shot or few-shot learning approach by including example data in the prompt.

A powerful tool is to show LLMs via examples within a prompt how to go about and format output.

A static prompt is just that, plain text with no templating, injection or external information.

https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350

Prompt Templating

Prompt Templating

Prompt templating is where a static prompt is converted into a template with key values being replaced with placeholders. The placeholders are replaced with application values/variables at runtime.

Some refer to templating as entity injection or prompt injection.

Prompt templating allows for prompts to the *stored*, *re-used*, *shared*, and *programmed*. Generative prompts can be incorporated into programs for programming, storage and re-use.

https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350

Prompt Templating

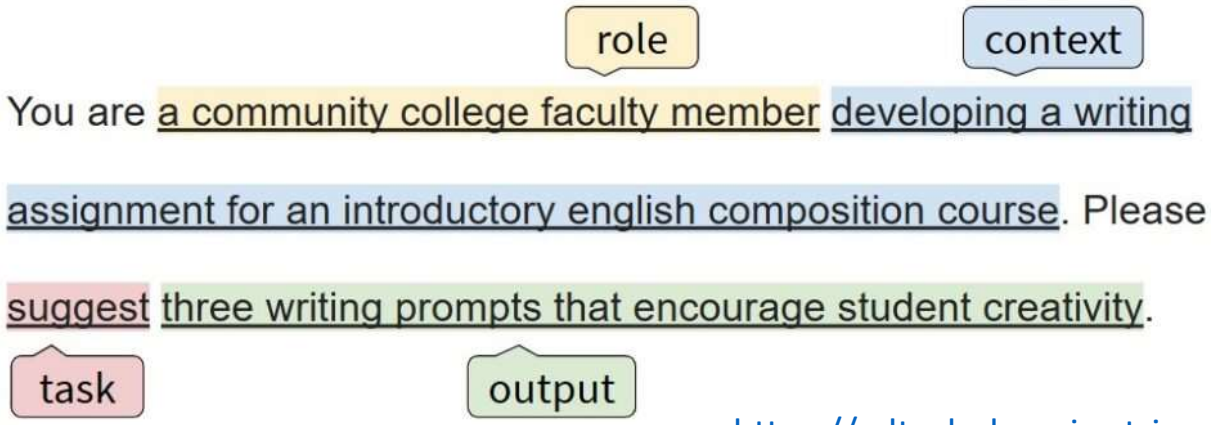
Technique	Prompt template	Description
1. Role Playing	"You are an expert in {field} known for {key adjective}. Help me {task}."	Telling the AI to act as a famous expert or celebrity
2. Style Unbundling	"Describe the key elements of {expert}'s style/skill in bullet points." "Do {task} in the following style: {style}."	Describe what you like about a style rather than copying directly
3. Emotion Prompting	"Help me {task}. Please make sure {attribute}. This task is very important for my career."	Use emotional blackmail and persuasion with the AI
4. Few-Shot Learning	"Here are some examples of {task}. Generate a {task} for {new context}."	Adding examples of the completed task to the prompt

Prompt Composition

Prompt Composition

A next step is to have a library of prompt templates which are combined at run-time to create a more advanced prompt. As prompt composition add a level of flexibility and programmability, it also introduces quite a bit of complexity.

https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350



https://edtech.domains.trincoll.edu/prompting-with-ai/

Contextual Prompts

Contextual Prompts

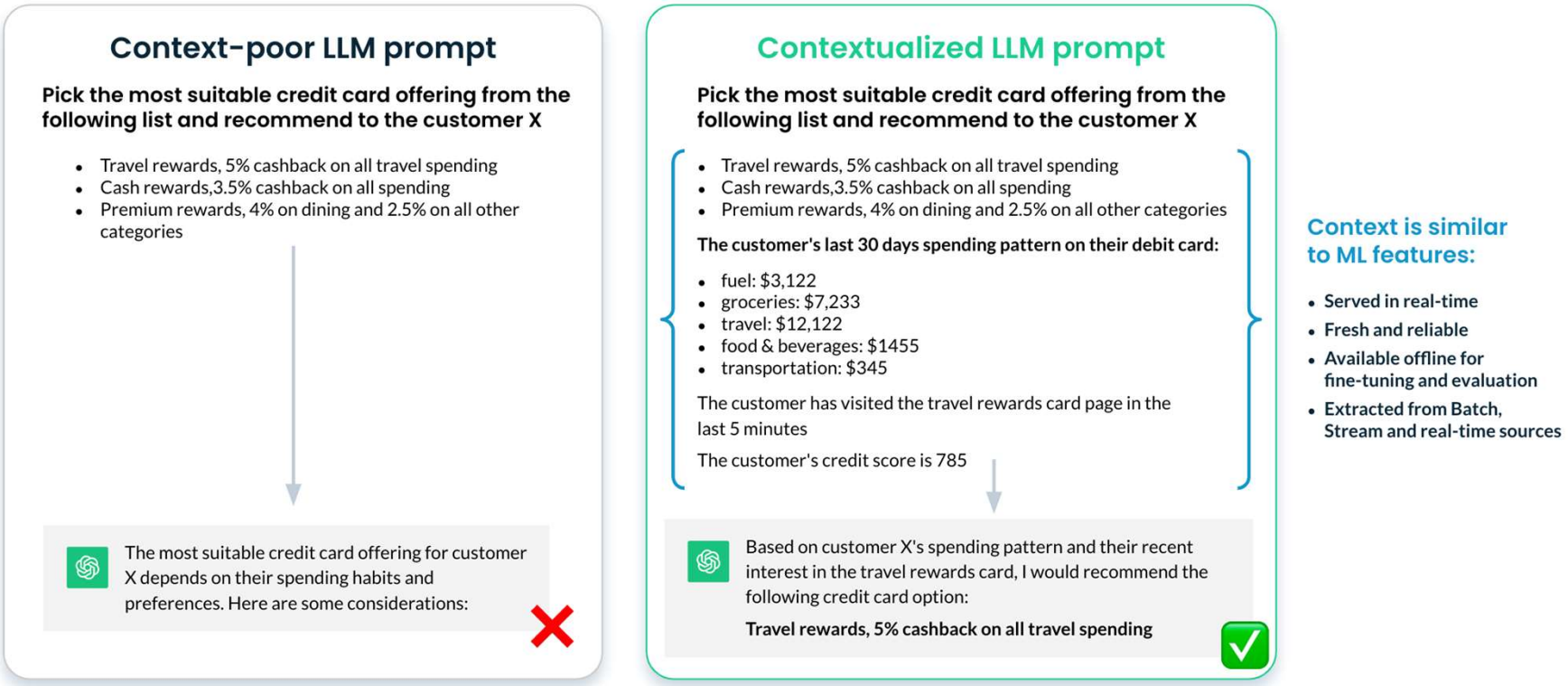
Even for LLMs, context is very important for increased accuracy and solving for hallucination. little context can go a long way in improving the accuracy of engineered prompts.

Contextual prompting negates LLM hallucination to a large extent.

Contextual prompting provides a frame of reference to the LLM when a response is generated.

https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350

Contextual Prompts



https://docs.tecton.ai/docs/0.9/introduction/llm-prompt-engineering

Prompt Chaining

Prompt Chaining

Prompt *Chaining,* also referred to as Large Language Model (LLM) *Chaining,* is the notion of creating a chain consisting of a series of model calls. This series of calls follow on each other with the output of one node in the chain serving as the input of the following.

Each chain node is intended to target small and well scoped sub-tasks, hence one or more LLMs is used to address multiple sequenced sub-components of a task.

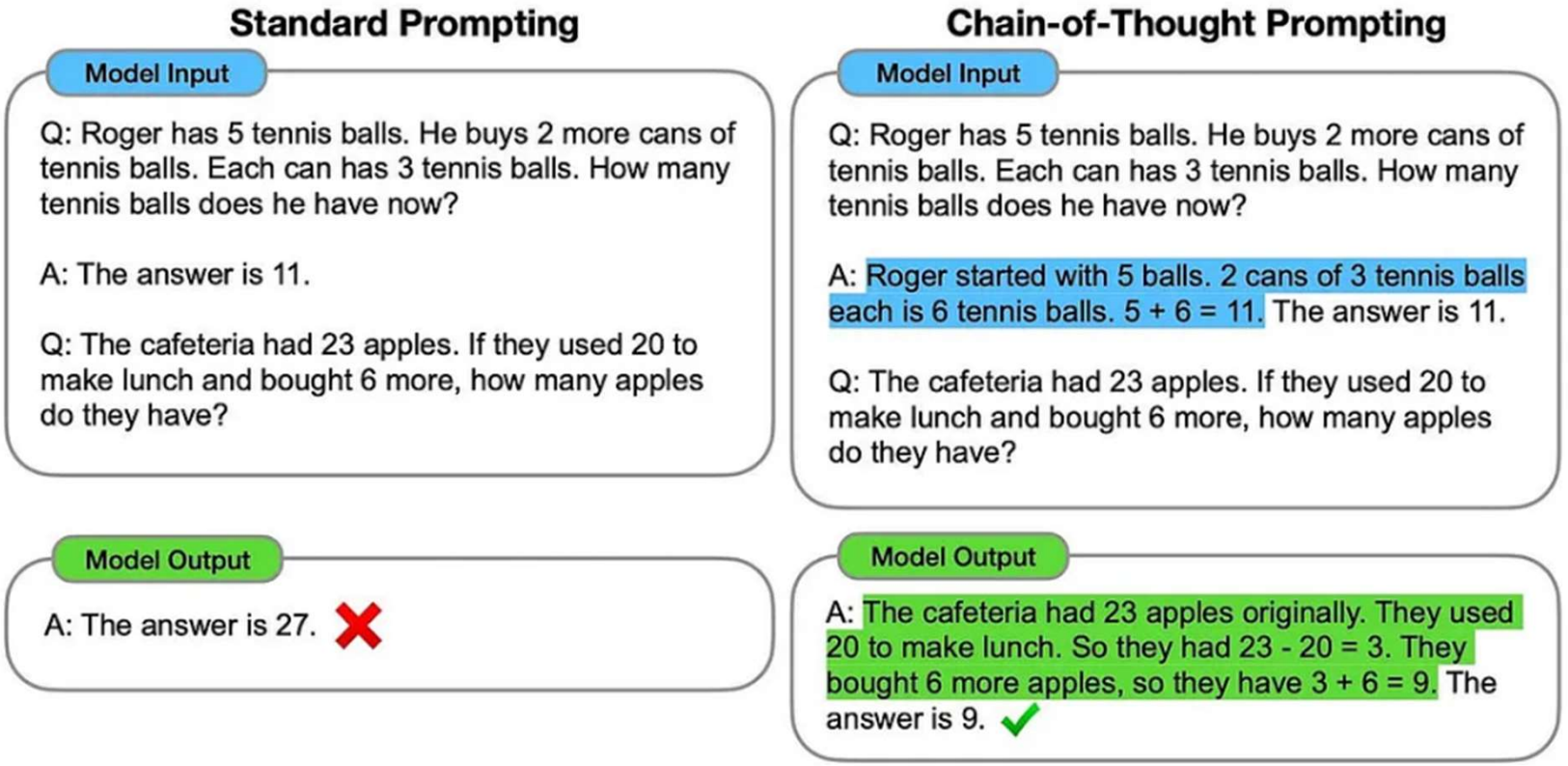
In essence prompt chaining leverages a key principle in prompt engineering, known as chain of thought prompting.

The principle of Chain of Thought prompting is not only used in chaining, but also in Agents and Prompt Engineering.

Chain of thought prompting is the notion of decomposing a complex task into refined smaller tasks, building up to the final answer.

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Example of Prompt Chaining



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Prompt Pipelines

Prompt Pipelines

In Machine Learning a pipeline can be described as and end-to-end construct, which orchestrates a flow of events and data.

The pipeline is kicked-off or initiated by a trigger; and based on certain events and parameters, a flow is followed which results in an output.

In the case of a prompt pipeline, the flow is in most cases initiated by a user request. The request is directed to a specific *prompt template*.

Prompt Pipelines can also be described as an intelligent extension to prompt templates.

The variables or placeholders in the pre-defined prompt template are populated (also known as prompt injection) with the question from the user, and the knowledge to be searched from the knowledge store.

This process can be highly dynamic and autonomous, crafting the desired prompt with the given scenario containing contextual information for few-shot training.

https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350

Prompt Tuning / Soft Prompts

Prompt Tuning / Soft Prompts

Soft prompts are created during the process of prompt tuning.

Unlike hard prompts, soft prompts cannot be viewed and edited in text. Prompts consist of an embedding, a string of numbers, that derives knowledge from the larger model.

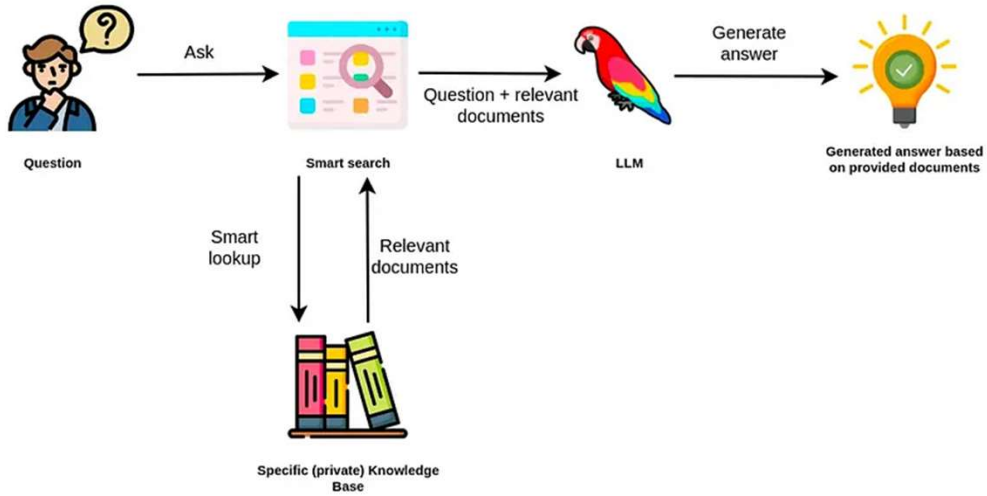
So for sure, a disadvantage is the lack of interpretability of soft prompts. The AI discovers prompts relevant for a specific task but can't explain why it chose those embeddings. Like deep learning models themselves, soft prompts are opaque.

Soft prompts act as a substitute for additional training data. Researchers recently estimated that a good language classifier prompt is worth hundreds to thousands of extra data points.

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Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI technique that combines a large language model (LLM) with an external knowledge base to improve response accuracy and relevance. Before generating an answer, RAG first retrieves relevant information from sources like internal documents or the internet to supplement the LLM's training data. This allows LLMs to produce more specific, up-to-date, and grounded responses without needing to be retrained.

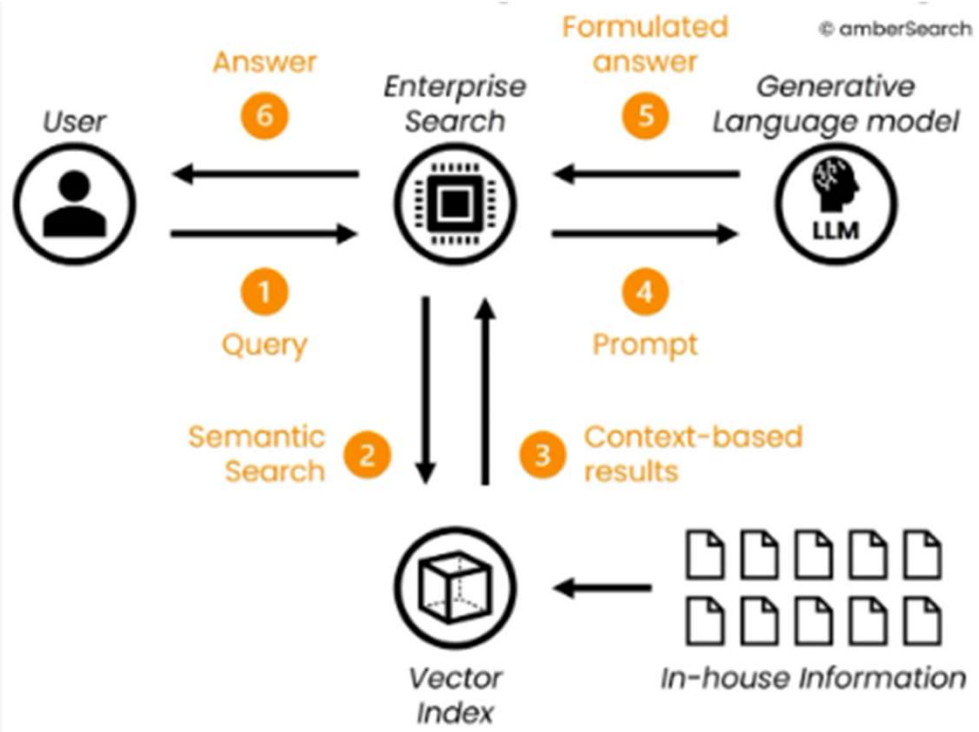


https://neo4j.com/blog/developer/rag-tutorial/

Retrieval Augmented Generation (RAG)

How RAG works:

- Retrieval: When a user asks a question, an external system first searches a knowledge base (like a vector database or document repository) to find information that is most relevant to the query.
- Augmentation: The retrieved information is then added to the original user prompt to create an "augmented prompt".
- Generation: The LLM receives this augmented prompt and generates a response that is based on both its pre-existing knowledge and the newly provided, external context.



https://ambersearch.de/en/what-is-retrieval-augmented-generation/

Retrieval Augmented Generation (RAG)

Key benefits:

- Improved accuracy: RAG grounds responses in reliable, external data, which minimizes the chance of the LLM "hallucinating" or providing incorrect information.
- Timeliness: It provides access to the most current information, as the knowledge base can be updated independently of the LLM.
- Cost-effective: It allows organizations to adapt LLMs to specific domains or internal data without the high cost and effort of full model retraining.
- Context-specific answers: RAG enables the model to give more specific and relevant answers by pulling from a specialized or internal knowledge source.
- Traceability: RAG can provide citations for its answers by linking them back to the original source documents

Retrieval Augmented Generation (RAG) – Limitations

- **Neglecting Relationships**: In practice, textual content is not isolated but interconnected. RAG fails to capture significant structured relational knowledge that cannot be represented through semantic similarity alone. Traditional systems don't effectively capture how different parts of the knowledge base relate to each other, leading to incomplete or disconnected information retrieval.
- **Redundant Information**: RAG often recounts content in the form of textual snippets when concatenated as prompts. This makes the context excessively lengthy, leading to the "lost in the middle"dilemma.
- **Lacking Global Information**: RAG can only retrieve a subset of documents and fails to grasp global information comprehensively. By splitting knowledge bases into small chunks during indexing, these systems often miss the bigger picture. Imagine trying to understand a novel by only reading random paragraphs - you'd miss crucial plot connections and themes.

GraphRAG

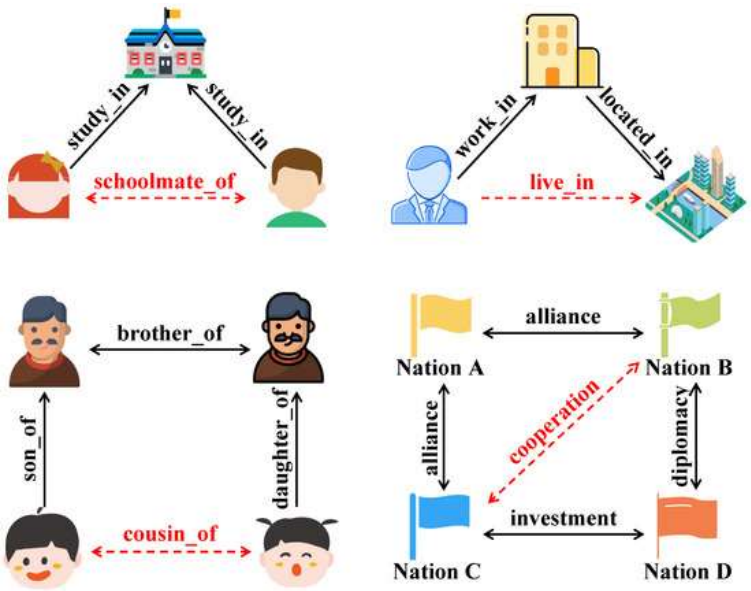
GraphRAG addresses traditional RAG limitations by using graph structures to represent knowledge, capturing complex relationships between entities and concepts.

Unlike flat document-based approaches, GraphRAG represents hierarchies, associations, and multi-hop relationships, providing richer semantic context for queries.

- **Enhanced Knowledge Representation**: Using graph structures to capture complex relationships
- **Multi-Source Integration**: Combining structured and unstructured data
- **Improved Efficiency**: Faster responses with fewer tokens
- **Better Interpretability**: Visualizing reasoning paths through knowledge

GraphRAG is good for Explainability

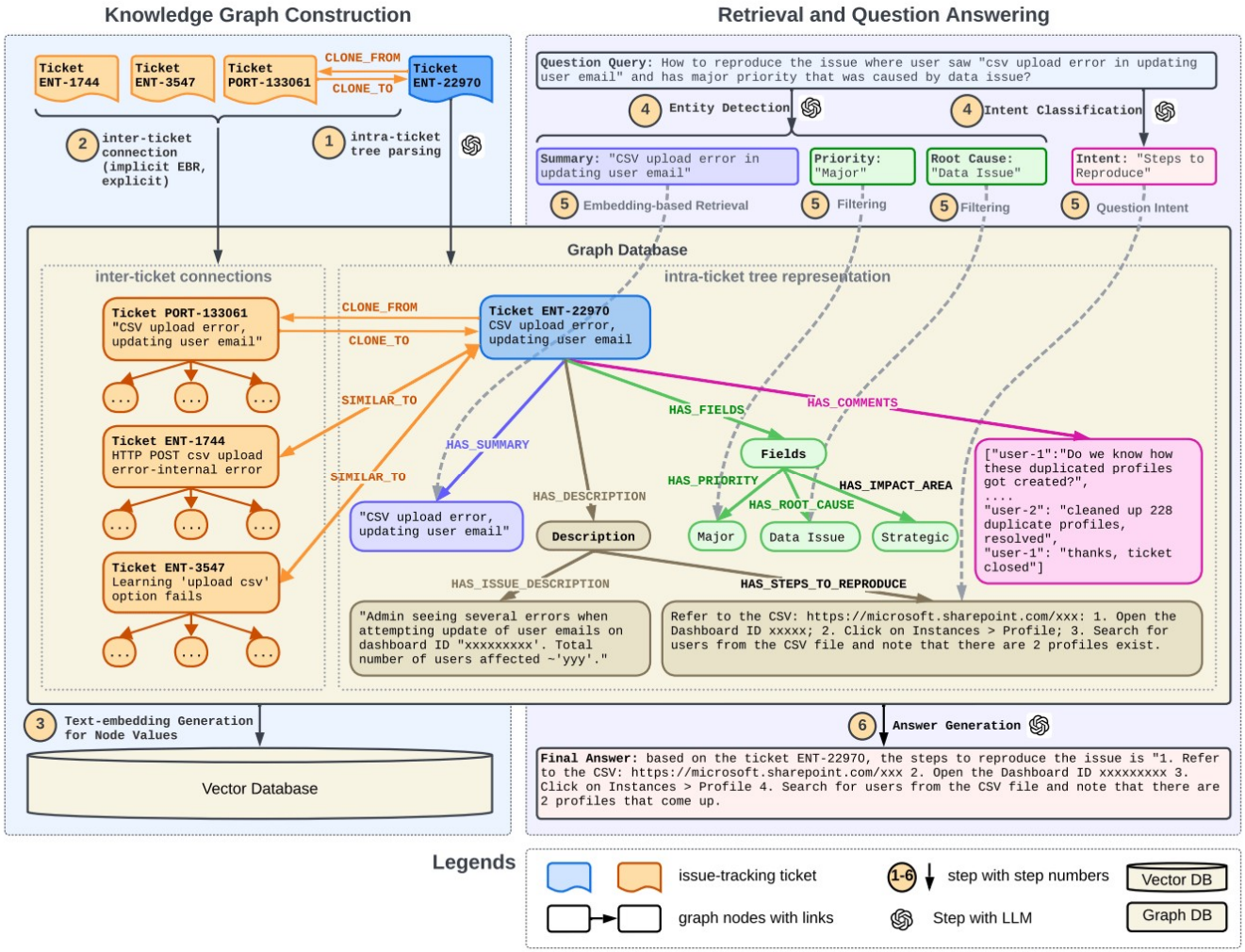
Examples of semantic context



https://www.mdpi.com/2227-7390/12/3/450

Case Study

Online professional platform LinkedIn introduces a novel customer service question-answering method that combines RAG with a Knowledge Graph



https://www.evidentlyai.com/blog/rag-examples

Autonomous Agents

Autonomous Agents

With LLM related operations there is an obvious need for automation. Currently this automation is in the form of what is called agents.

Prompt Chaining is the execution of a predetermined and set sequence of actions.

The attraction of Agents is that Agents do not follow a predetermined sequence of events. Agents can maintain a high level of autonomy.

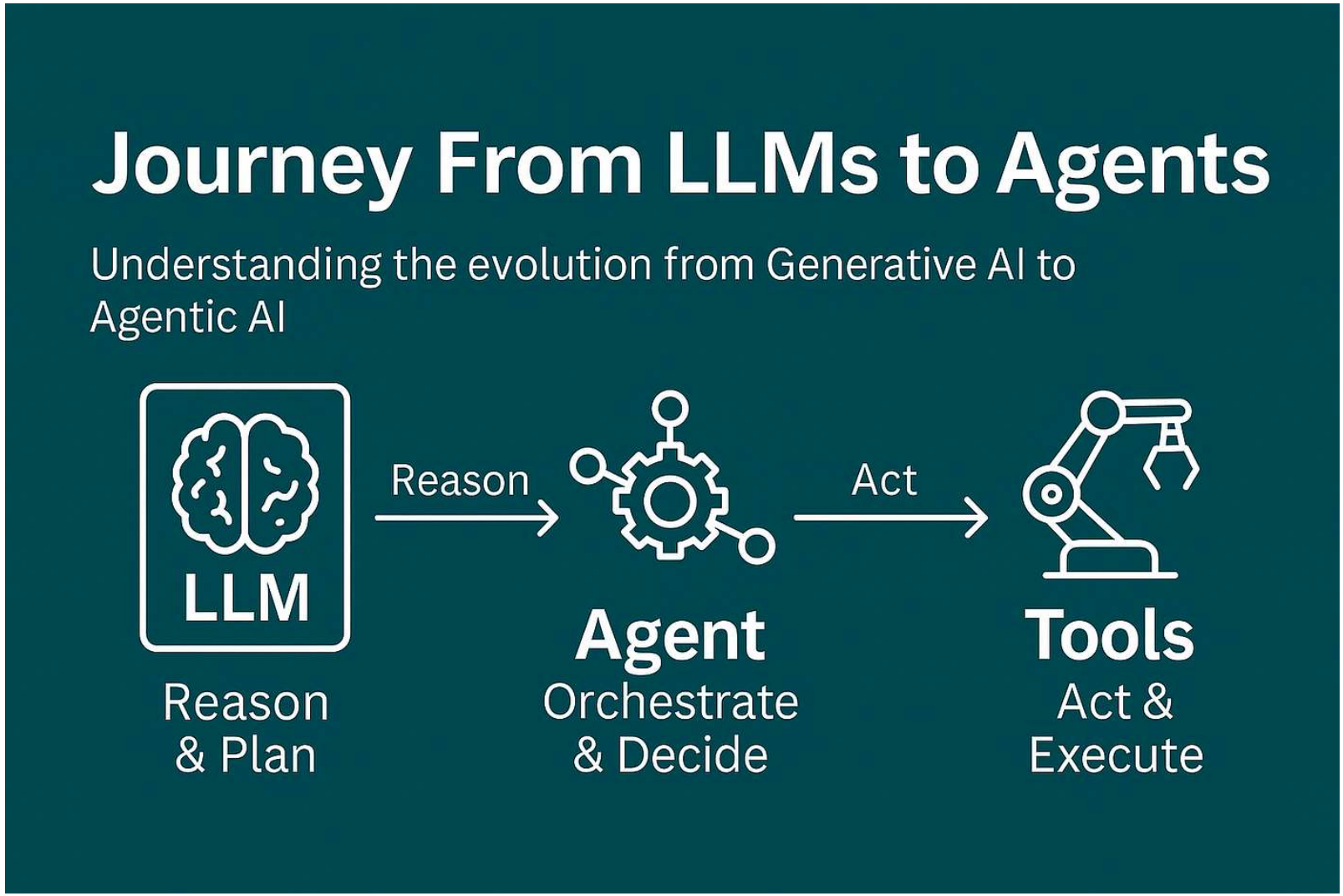
Agents have access to a set of tools and any request which falls within the ambit of these tools can be addressed by the agent. The Execution pipeline lends autonomy to the Agent and a number of iterations might be required until the Agent reaches the Final Answer.

https://cobusgreyling.medium.com/eight-prompt-engineering-implementations-updated-90c82d071350

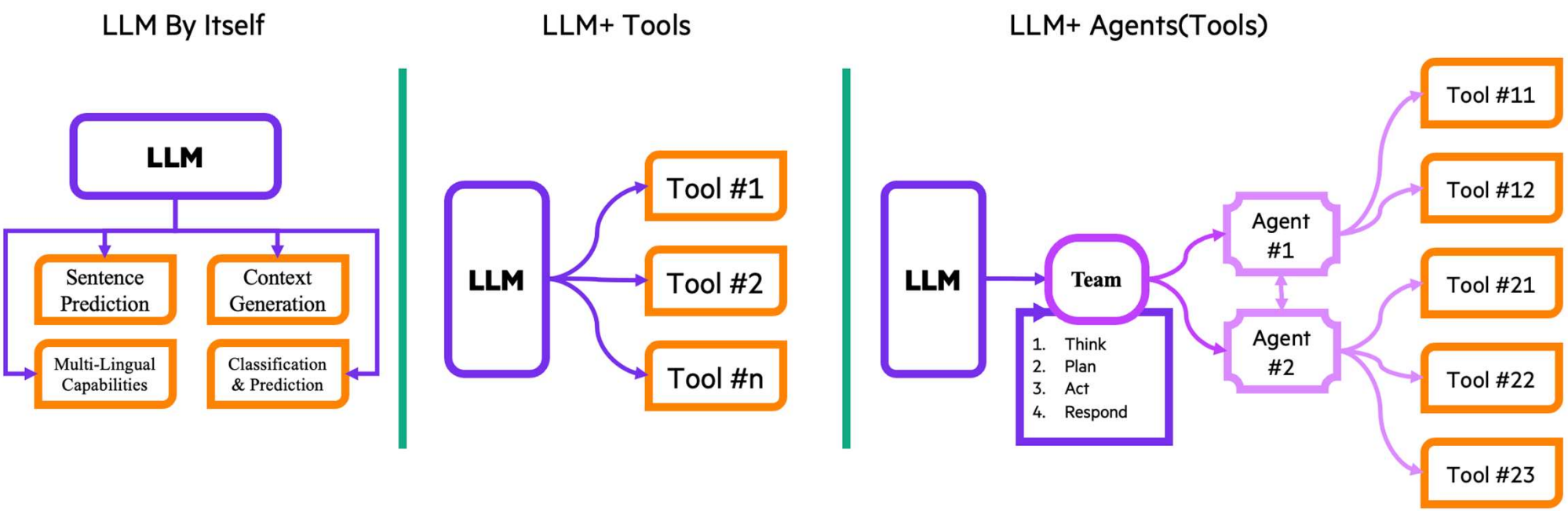
Agentic AI

Agentic AI builds on Generative AI (GenAI) techniques by using large language models (LLMs) to function in dynamic environments. While generative models focus on creating content based on learned patterns, Agentic AI extends this capability by applying generative outputs toward specific goals. A generative AI model like OpenAI's ChatGPT might produce text, images or code, but an agentic AI system can use that generated content to complete complex tasks autonomously by calling external tools

IBM

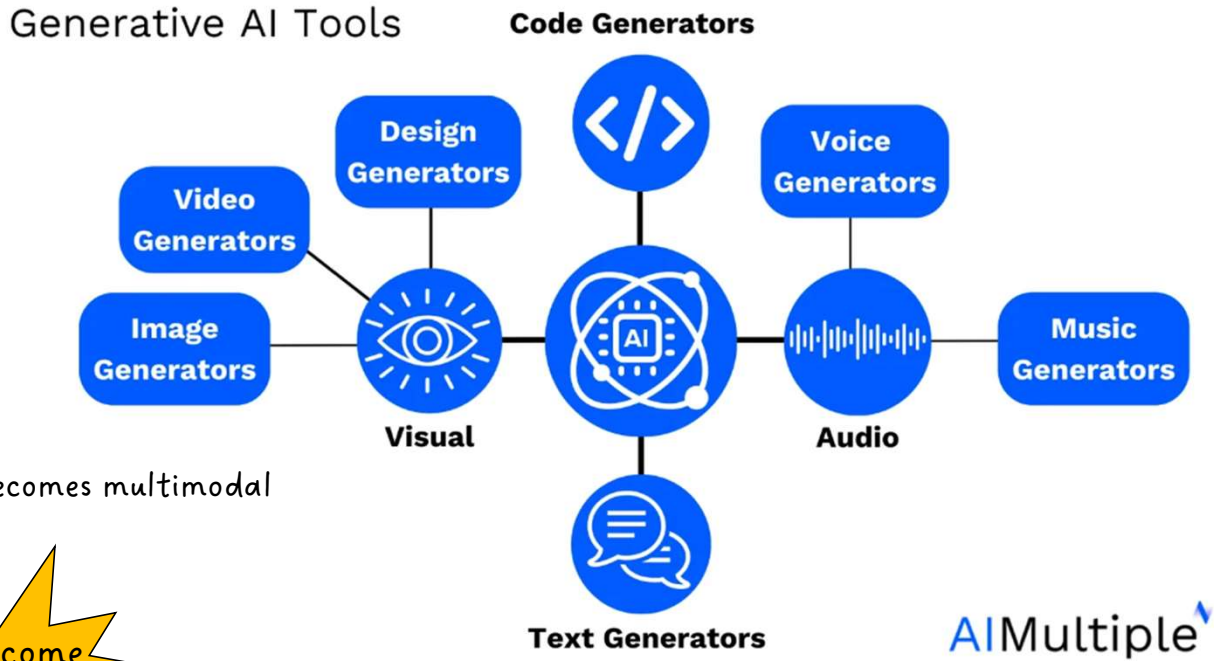


https://medium.com/@khannap2001/your-journey-from-llms-to-agents-6e9d143d2b73



https://developer.hpe.com/blog/from-generative-to-agentic-ai-tracing-the-leap-from-words-to-actions/

Generative AI Tools

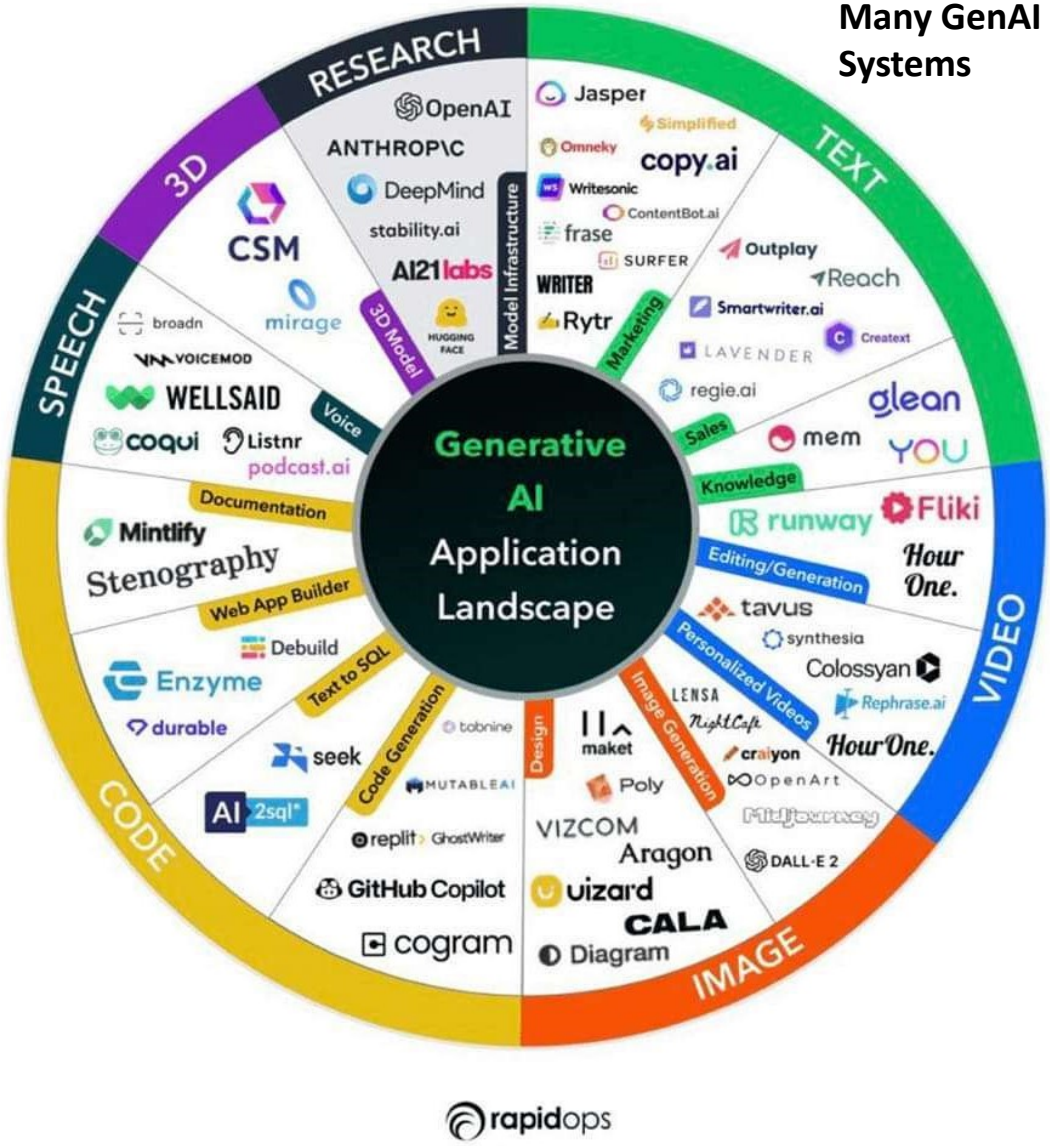


Generative AI becomes multimodal



Generative Ai Tools: https://research.aimultiple.com/generative-ai-tools/

Many GenAI Systems



rapidops

AI TOOLS YOU NEED TO KNOW

Create by Rene Remsik @aitrendz.xyz

AI Presentation
- Beautiful.AI
- Gamma
- Pitch
- Plus
- PopAI
- Presentation.Ai
- Slidesgo
- Tome

AI Chatbots
- ChatGPT
- Claude
- DeepSeek
- Gemini
- Grok
- Meta AI
- MS Copilot
- Perplexity

AI Email Assistance
- Clippit.Ai
- Friday
- Mailmaestro
- Shortwave
- Superhuman

AI Coding Assistance
- Askcodi
- Codiga
- Cursor
- GitHub Copilot
- Qodo
- Replit
- Tabnine

AI Apreadsheet
- Bricks
- Formula Bot
- Gigasheet
- Rows AI
- SheetAI

AI Image Generation
- Adobe
- DALL-E
- FLUX.1
- Ideogram
- Midjourney
- Recraft
- Stable Diffusion

AI Workflow Automation
- Integrately
- Make
- Monday.com
- n8n
- Wrike
- Zapier

AI Meeting Notes
- Avoma
- Equal Time
- Fathom
- Fellow.app
- Fireflies
- Krisp
- Otter

AI Scheduling
- Calendly
- Clockwise
- Motion
- Reclaim AI
- Taskade
- Trevor AI

AI Writing Generation
- Copy.ai
- Grammerly
- Jasper
- JotBot
- Quarkle
- Quillbot
- Rytr
- Sudowrite
- Writesonic

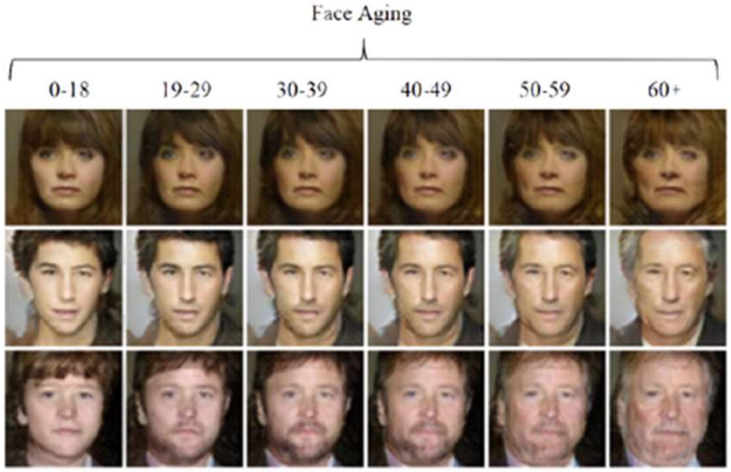
AI Graphic Design
- AutoDraw
- Canva
- Design.com
- Framer
- Microsoft Designer
- Uizard

AI Video Generation
- Descript
- Haiper AI
- Invideo AI
- Kling
- Krea AI
- LTX Studio
- Luma AI
- Pika AI
- Runway
- Sora

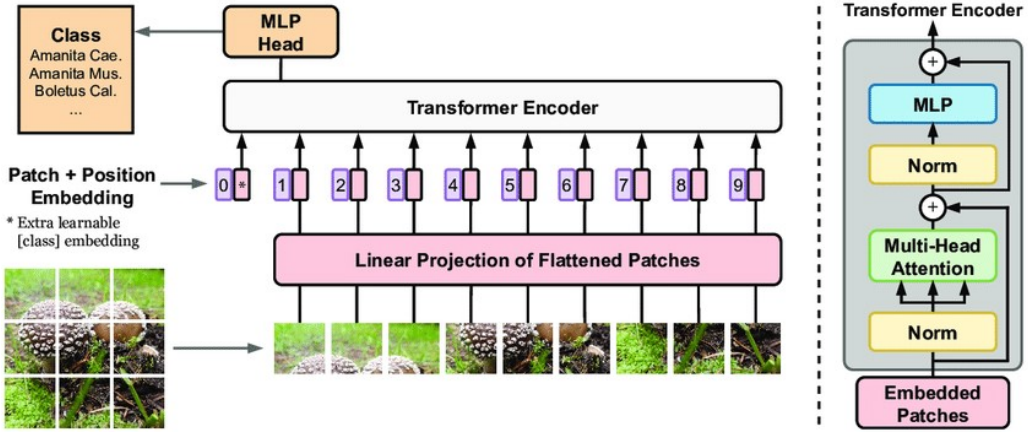
AI Knowledge Management
- Mem
- Notion
- Tettra

AI Data Visualization
- Deckpilot
- Flourish
- Julius
- Visme
- Zing Data

Image/Video Generation



"Face Aging With Conditional Generative Adversarial Networks".



Visual Transformers (ViT)





Visual Transformers (ViT)

Vision Transformers (ViTs) have emerged as a competitive alternative to convolutional neural networks (CNNs), which were previously the state of the art in computer vision and widely used for various image recognition tasks

Although convolutional neural networks have dominated the field of computer vision for years, new vision transformer models have also shown remarkable capabilities, achieving comparable and even better performance than CNNs in many computer vision tasks

ViT models outperform current state-of-the-art CNNs by almost four times in terms of computational efficiency and accuracy

However, ViTs require large volumes of data, so in the absence of this amount of data, CNNs may operate better

For the same amount of data, ViTs require less training time.

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

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ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.[1]

Oririnal scientific paper about the application of Transformers to Image Recognition (https://arxiv.org/pdf/2010.11929)

Visual Transformers (ViT)

Let's examine the vision transformer architecture step by step:

1. Divide an image into patches
2. Flatten the patches
3. Produce smaller-dimensional linear embeddings from the flattened patches
4. Add positional embeddings
5. Feed the sequence as input to a standard transformer encoder
6. Pre-train the model with image labels (fully supervised on a huge dataset)
7. Fine-tune the downstream dataset for image classification

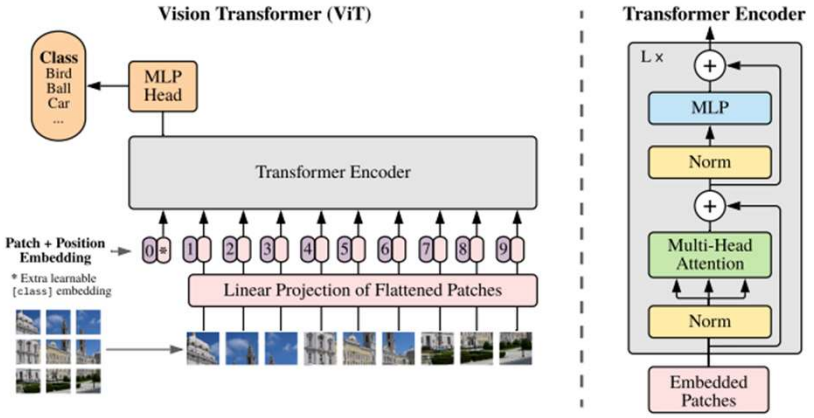


Figure 1: Model overview. We split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable "classification token" to the sequence. The illustration of the Transformer encoder was inspired by Vaswani et al. (2017).

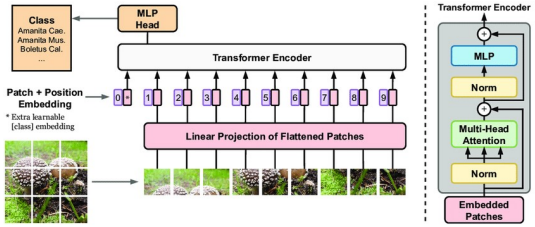
Note: a 28x28 image has 784 pixels. The mechanism of Attention of the transformers would be a 784x784 matrix. By dividing it into patches, we will have a much lower dimensionality.

Visual Transformers (ViT)

Image patches are sequence tokens (like words)

The encoder block is identical to the original transformer proposed by Vaswani et al. (2017)

There are several blocks in the ViT encoder, and each block consists of three main processing elements:



- Layer Norm
- Multi-Head Attention Network
- Multilayer Perceptrons (MLP)

The Layer Norm keeps the training process on track and allows the model to adapt to variations between training images

The Multi-Head Attention Network is a network responsible for generating attention maps from embedded visual tokens. These attention maps help the network focus on the most critical regions of the image, such as objects. The concept of attention maps is the same as that found in traditional computer vision literature

The MLP is a two-layer classification network with GELU (Gaussian Error Linear Unit) at the end. The final MLP block, also called the MLP head, is used as the transformer output. A softmax application on this output can provide classification labels (for example, if the application is Image Classification)

Visual Transformers (ViT)

ViT versus CNN

1. ViTs have greater similarity between representations obtained in shallow and deep layers compared to CNNs
2. Unlike CNNs, ViTs obtain the global representation of shallow layers, but the local representation obtained from shallow layers is also important
3. The skipped links in ViTs are even more influential than in CNNs (ResNet) and substantially impact the performance and similarity of the representations
4. ViTs retain more spatial information than ResNet
5. ViTs can learn high-quality intermediate representations with large amounts of data
6. The MLP-Mixer representation is closer to ViTs than to ResNet

Visual Transformers (ViT)

What ViT enables:

- Image Classification
- Image Captioning
- Image Segmentation
- Anomaly Detection
- Action Recognition
- Autonomous Driving Support

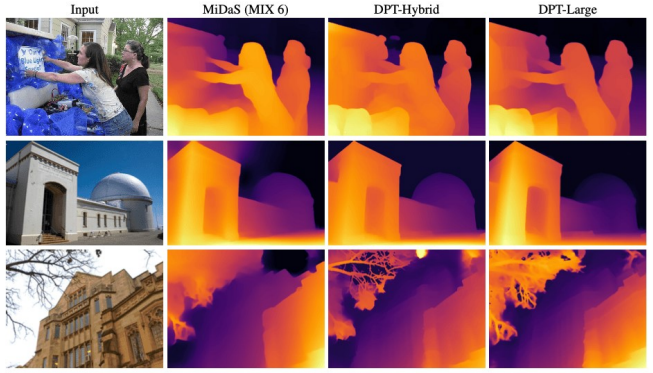
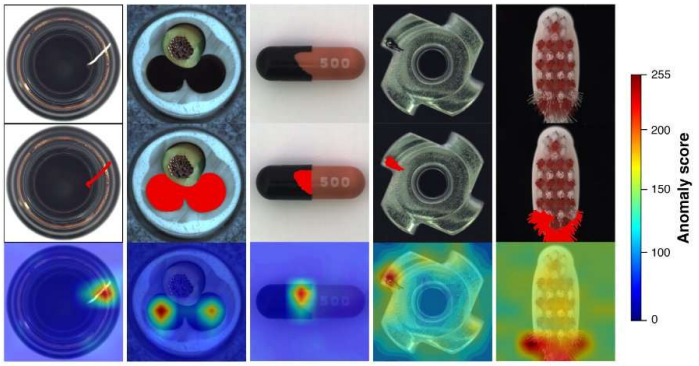
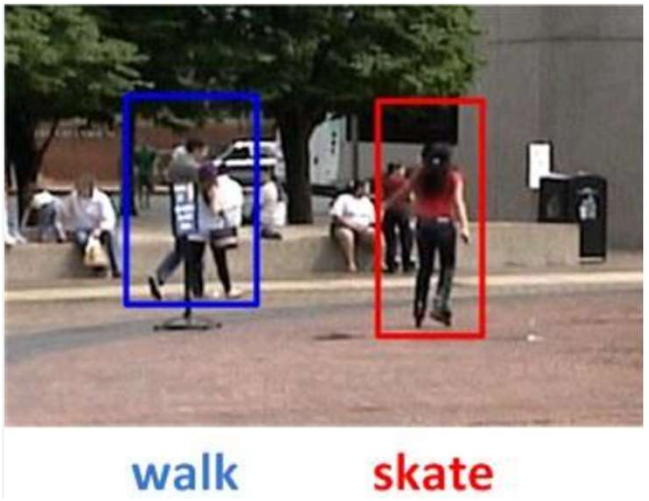
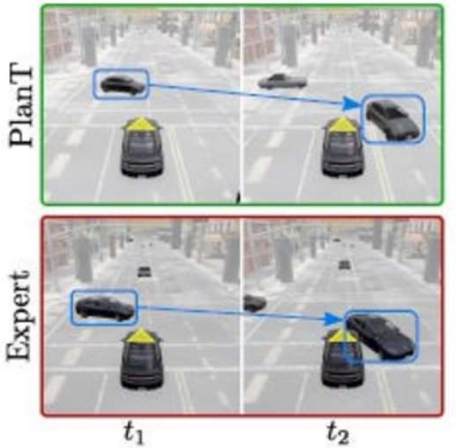


Figure 2. Sample results for monocular depth estimation. Compared to the fully-convolutional network used by MiDaS, DPT shows better global coherence (e.g., sky, second row) and finer-grained details (e.g., tree branches, last row).







(a) **PlanT vs. Expert.** PlanT waits further outside the intersection than the expert to avoid a collision.

Image Generation

Current image generators are based on text-to-image conversion and use Deep Learning approaches, such as Visual Transformers.

Some examples:

Midjourney - for artistic results

DALL·E 3 - for incorporating AI images into your existing workflows

Ideogram - for precise texto

Stable Diffusion - for customization and control of your AI images

FLUX.1 - for a stable diffusion alternative

Adobe Firefly - for integrating AI-generated images into photos

Recraft - for graphic design

The best AI image generators at a glance

	Best for	Access options	Price	Parent company
Midjourney	Artistic results	Web app; Discord	From $10/month for ~200 images/month and commercial usage rights	Midjourney
DALLE-3	Integrating AI images in your existing workflows	ChatGPT Plus or Enterprise; Microsoft Copilot; API	2 free images/day with a free ChatGPT plan; included with ChatGPT Plus at $20/month	OpenAI
Ideogram	Accurate text	Web app	Limited free plan; from $8/month for full-resolution download and 400 monthly priority credits	Ideogram AI
Stable Diffusion	Customization and control	NightCafe, Tensor.Art, Civitai, and lots of other apps; API; downloading it to a local server	Depends on the platform	Stability AI
FLUX.1	Stable Diffusion alternative	NightCafe, Tensor.Art, Civitai, and lots of other apps; API; downloading it to a local server	Depends on the platform	Black Forest Labs
Adobe Firefly	Using AI-generated images in photos	firefly.adobe.com, Photoshop, Express, and other Adobe tools	Limited free credits; from $9.99 for 2,000 credits/month	Adobe
Recraft	Graphic design	Web app	Free for 50 credits/day; from $12/month for full features	Recraft

Video Generation

Sora - for amazing results with simple tips

Runway - for generative AI video with advanced tools

Descript - for video editing by editing the script

Wondershare Filmora - for polishing videos with AI tools

Capsule - for simplifying video production workflows with AI

invideo AI - for social media vídeos

Peech for content marketing teams

Synthesia - for using digital avatars

Vyond - for animated character videos from a prompt

revid.ai - for AI-powered templates

LTX Studio - for extreme creative control

The best AI video generators at a glance

	Best for	Platforms	Free plan
Sora	Amazing videos with simple prompts	Web	No (but it's part of ChatGPT Plus)
Runway	Generative AI video	Web, iOS	Yes, with 125 video credits (used for AI features)
Descript	Editing video by editing the script	Web, Windows, Mac	Yes, with 1 hour of transcription and watermark
Wondershare Filmora	Polishing video with AI tools	Windows, Mac, iOS, Android	Yes, with watermark
Capsule	Simplifying video production workflows with AI	Web (prefers Chrome)	Yes, for standard visual components and basic editing features
invideo AI	Social media videos	Web	Yes, for 10 minutes of generated video per week and 4 exports with watermark
Peech	Content marketing teams	Web (prefers Chrome)	Yes, for 1 user, 2 videos per month, with watermark
Synthesia	Using digital avatars	Web	Yes, for 9 avatars and 3 minutes per month
Vyond	Creating animated characters from a prompt	Web	No
revid.ai	AI-powered templates	Web	Yes, a suite of free AI video tools available
LTX Studio	Extreme creative control	Web	Yes, for personal use, 5 collaborators per project, and up to 1,000 computing seconds

Video Generators

Tool/Max Resolution	Free Trial	Cheapest Price (USD)	Text to Video	Image to Video	Edit/Update Output	Max Shot Length (sec)	Camera Controls	Lip Sync	Sound Generation
Kling AI/1080p	Y	$10/month	Y	Y	Y	10	Y	Y	N
Runway Gen 4/1080p	Y	$12/month	Y	Y	Y	16	Y	Y	N
Google Veo 2/4K	Y	$30/min ($0.50/sec)	Y	N	Y	120	Y	N	N
OpenAI Sora/1080p	N	$20/month (Plus) $200/month (Pro)	Y	Y	Y	20 (Pro); 5 (Plus)	N	N	N
Pika 2.0/1080p	Y	$35/month	Y	Y	Y	16	N	Y	N
Adobe Firefly/1080p	Y	$9.99 month	Y	Y	Y	5	Y	N	N
Hailuo AI/1080p	Y	$14.99 month	Y	Y	Y	5 (free)	N	Y	N
Luma Dream Machine/1080p	Y	$9.99 month	Y	N	N	10	N	N	Y
Artlist/1080p	N	$29.99/month	Y	Y	Y	10	N	N	N
Vidu/1080p	Y	$8/month	Y	Y	Y	5	Y	Y	Y